Chapter 2: Machine Learning

Machine Learning (ML) is a branch of AI in which computer systems learn patterns from data instead of being programmed with every rule. In supervised learning, a model learns from labeled examples. Classification predicts categories, while regression predicts numerical values. Common supervised learning algorithms include linear regression, decision trees, support vector machines, and neural networks.

Unsupervised learning works with data that does not have labeled answers. Clustering is a common unsupervised learning task in which similar observations are grouped together. A machine learning workflow commonly includes collecting data, cleaning data, splitting data into training and testing sets, training a model, evaluating performance, and improving the model.